

Notes on “The Financial Premium”

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For all figures and tables referenced in these notes, see the working paper at:
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1 Executive Takeaways

- **Empirical fact:** Corporate bonds issued by financial firms trade at higher yield spreads than bonds issued by industrial firms with the *same rating and maturity*. The difference is the *financial premium*.
- **Magnitude:** In U.S. data 1987–2020, the average premium is about $\approx 43\text{bp}$, larger for lower ratings, larger in financial crises, and strongly related to bond β .
- **Theory:** Financial firms (banks, insurers, financing companies) hold *large diversified portfolios* of non-financial debt. Diversification removes idiosyncratic risk on the asset side, so their liabilities load more on *systematic* risk than a standalone non-financial firm’s debt with the same default probability. Hence a positive premium arises as compensation for systematic risk.
- **Key model prediction:** Financial bonds have higher “excess debt beta” (beta difference relative to comparable industrial bonds), and the premium is increasing in that beta.
- **Robustness:** Differences in default/loss rates, transaction-based liquidity, covenants, large-issuer demand effects, and slow rating updates do not explain away the premium.

2 Empirical Object: What is the “Financial Premium”?

2.1 Definition in words

The **financial premium** is the extra yield spread (over a risk-free benchmark) paid by bonds issued by *financial* firms relative to *industrial* firms, after controlling finely for rating and maturity (and optionally other bond characteristics).

2.2 Baseline identification regression (panel)

The paper operationalizes the premium by estimating a panel regression of bond-month yield spreads:

$$s_{itj} = \phi \mathbf{1}\{j \in \text{Financial}\} + \gamma' X_{it} + \mu_{mrt} + \varepsilon_{itj}, \quad (1/9)$$

where:

- s_{itj} is the yield spread (bond yield-to-maturity minus swap rate of comparable maturity) for bond i issued by firm j in month t ;
- $\mathbf{1}\{j \in \text{Financial}\}$ indicates whether issuer j is financial (vs industrial);
- X_{it} includes standard liquidity proxies: coupon, bond age, and $\log(\text{amount issued})$;

- μ_{mrt} is a *month* $\times$ *rating* $\times$ *maturity* fixed effect (rating at notch level; maturity in bins like 0.5–1.5, 1.5–2.5, ..., 9.5–10.5 years).

Interpretation: ϕ is the average spread difference between financial and industrial bonds that are matched on rating, maturity, and time (and on X_{it} if included). This is the financial premium.

2.3 Why controlling for rating and maturity is essential

Financial firms tend to issue:

- *higher-rated* bonds, and
- *shorter-maturity* bonds,

relative to industrial firms. Because spreads vary strongly with rating and typically with maturity, raw average spreads can be misleading. Matching within month $\times$ rating $\times$ maturity cells makes the premium interpretable as a within-cell difference.

3 Theory: A model that generates a positive financial premium

3.1 High-level idea

A non-financial firm is exposed to both systematic and idiosyncratic shocks. Its debt is therefore not purely a claim on the market factor. A financial firm that holds a *large diversified portfolio* of many non-financial debts effectively diversifies away idiosyncratic risk and ends up holding (almost) only systematic risk. As a result, *for the same default probability*, the financial firm's debt is more exposed to systematic risk and commands a higher risk premium.

3.2 The model primitives

Market portfolio. Under the physical measure $\mathbb{P}$, the market portfolio evolves as

$$dV_m(t) = \mu_m V_m(t)dt + \sigma_m V_m(t)dW_m(t), \quad (2)$$

with constant risk-free rate r . The market Sharpe ratio is

$$s_m = \frac{\mu_m - r}{\sigma_m} > 0.$$

Non-financial firm asset values (Merton-style). There are N identical firms with asset values

$$dV_i(t) = \mu V_i(t)dt + V_i(t)(\beta\sigma_m dW_m(t) + \nu dW_i(t)), \quad (3)$$

where dW_i is idiosyncratic and independent across firms and independent of dW_m . Total asset volatility:

$$\sigma = \sqrt{(\beta\sigma_m)^2 + \nu^2}, \quad \rho = \frac{(\beta\sigma_m)^2}{\sigma^2}$$

so ρ is the pairwise correlation between firms' log asset values.

Debt contracts. Each firm issues zero-coupon debt with maturity T , face value D_f , and exogenous recovery rate R_f . Default occurs at maturity if $V_i(T) < D_f$.

Financial firm (“bank”) balance sheet. A representative financial firm holds a *large diversified portfolio* of the firms' debts. The key asymptotic step is a Vasicek-style large- N approximation: as $N \rightarrow \infty$, idiosyncratic components wash out and the portfolio loss becomes a function of the common market factor only.

The financial firm issues its own zero-coupon debt with maturity T , face D_b , and recovery rate R_b . It chooses leverage such that its debt has default probability p_b .

3.3 Pricing logic: CAPM and risk-neutral default probabilities

The drift μ of a firm asset is pinned down by CAPM:

$$\mu - r = \beta(\mu_m - r).$$

Therefore the firm's asset Sharpe ratio is

$$s_f \equiv \frac{\mu - r}{\sigma} = \frac{\beta(\mu_m - r)}{\sigma} = \sqrt{\rho} s_m.$$

This mapping is what transforms a *physical* default probability p_f into a *risk-neutral* default probability $q_f(p_f)$.

4 Main theoretical results

4.1 Yield spreads for firm debt vs bank debt

Proposition 1 (Credit spreads of firm and bank debt). *At time 0, the yield spread (yield-to-maturity minus the risk-free rate) of:*

1. *firm debt is*

$$y_f(p_f, R_f, T) = -\frac{1}{T} \log \left(1 - (1 - R_f) \Phi(\Phi^{-1}(p_f) + \sqrt{\rho} s_m \sqrt{T}) \right),$$

2. *bank debt is*

$$y_b(p_b, R_b, T) = -\frac{1}{T} \log \left(1 - (1 - R_b) \Phi(\Phi^{-1}(p_b) + s_m \sqrt{T}) \right),$$

where Φ is the standard normal CDF and N is large so that the Vasicek (1991) diversified-loan approximation holds for the bank asset portfolio.

Intuition.

- A standalone firm's asset Sharpe ratio is $\sqrt{\rho} s_m$ because it mixes systematic and idiosyncratic risk.
- A fully diversified bank debt portfolio has (essentially) *only systematic risk* and therefore inherits Sharpe ratio s_m .
- Since $s_m > \sqrt{\rho} s_m$, the bank debt has a larger “risk-neutral tilt” for a given physical default probability, which shows up as a higher spread.

4.2 Definition and properties of the financial premium

Definition 1 (Financial premium). For maturity T , default probability p , and recovery rate R , the **financial premium** is

$$f(p, R, T) \equiv y_b(p, R, T) - y_f(p, R, T). \quad (4)$$

Proposition 2 (Properties of the financial premium). *The financial premium $f(p, R, T)$ satisfies:*

1. $f(p, R, T) > 0$ (bank debt has a higher spread than firm debt with the same (p, R, T)).
2. $f(p, R, T)$ is increasing in the default probability p for a wide range of empirically relevant p (the paper provides an explicit sufficient condition).
3. $f(p, R, T)$ is increasing in the market Sharpe ratio s_m (more priced systematic risk implies a larger premium).

Economic interpretation. Financial debt is a *more systematic claim* than non-financial debt with the same default probability. Hence investors demand a higher expected return (a higher spread).

4.3 Bond betas and “excess debt beta”

The model delivers predictions not only for spreads but also for *bond betas*.

Proposition 3 (Betas of firm and bank debt). *Let the instantaneous return processes be*

$$dr_f(t) = \mu_f(t)dt + \sigma_{fm}(t)dW_m(t) + \sigma_{fi}(t)dW_i(t), \quad dr_b(t) = \mu_b(t)dt + \sigma_{bm}(t)dW_m(t).$$

Define debt beta as

$$\gamma_d(t) = \frac{\sigma_{dm}(t)}{\sigma_m}, \quad d \in \{f, b\}. \quad (5)$$

At time 0, the firm-debt and bank-debt betas are:

$$\gamma_f(p, R, T) = \frac{(1-R)\sqrt{\rho}}{\sigma_m\sqrt{T}} \frac{\varphi(\Phi^{-1}(p) + \sqrt{\rho}s_m\sqrt{T})}{1 - (1-R)\Phi(\Phi^{-1}(p) + \sqrt{\rho}s_m\sqrt{T})}, \quad (6)$$

$$\gamma_b(p, R, T) = \frac{(1-R)}{\sigma_m\sqrt{T}} \frac{\varphi(\Phi^{-1}(p) + s_m\sqrt{T})}{1 - (1-R)\Phi(\Phi^{-1}(p) + s_m\sqrt{T})}. \quad (7)$$

Definition 2 (Excess debt beta). The **excess debt beta** is the beta difference between bank debt and firm debt, holding (p, R, T) fixed:

$$\gamma^e(p, R, T) \equiv \gamma_b(p, R, T) - \gamma_f(p, R, T). \quad (8)$$

Proposition 4 (Excess debt beta is positive and (locally) increasing in p). *Under sufficient conditions provided in the paper, $\gamma^e(p, R, T)$ is:*

1. *strictly positive for a broad range of default probabilities p (including the empirically relevant region), and*
2. *increasing in p over an initial interval $(0, p^{++})$ (with explicit bounds in the appendix).*

Why betas matter for the empirical strategy. If the premium is a compensation for systematic risk, then across portfolios that differ in rating and maturity, we should observe:

$$\text{higher financial premium} \iff \text{higher excess debt beta}.$$

This is exactly what the paper tests in the data (Figure 3 and Table 6).

5 From Theory to Empirics: Measurement and Regression Models

5.1 Data construction (minimum needed facts)

The bond sample is U.S. corporate bonds from 1987–2020 assembled from standard sources (Lehman Brothers data for early years, Mergent FISD for issuance characteristics, TRACE for transactions in later years). The sample is restricted to senior unsecured fixed-rate (or zero-coupon) bonds, excludes special features (convertible/puttable/perpetual/etc.), excludes FDIC-guaranteed financial issuance in 2008–2009, and focuses on maturities 0.5–10.5 years.

Risk-free benchmark: swap rates (with interpolations where needed). Spreads are winsorized.

5.2 Main regressions (“models” in the empirical section)

(i) Financial premium. Equation (1/9) is the workhorse regression.

(ii) Bond betas and excess debt beta. Bond beta is computed for each bond-month via a time-series regression over the previous 36 months:

$$r_{i,\tau}^{\text{bond}} = a_i + \beta_{i,t}^{\text{bond}} r_{\tau}^{\text{mkt}} + u_{i,\tau}, \quad \tau = t - 36, \dots, t - 1,$$

where r^{bond} is the bond’s excess return and r^{mkt} is the equity market excess return. Then betas are compared across financial vs industrial using:

$$\beta_{itj} = \eta \mathbf{1}\{j \in \text{Financial}\} + \gamma' X_{it} + \mu_{mrt} + \varepsilon_{itj}. \quad (10)$$

The coefficient η is the empirical analogue of γ^e : a within-cell excess debt beta.

(iii) Premium–beta relation across rating/maturity buckets. For each rating r and maturity bucket m , define:

$$fpr_{r,m} = \text{estimated financial premium in } (r, m), \quad \beta_{r,m} = \text{estimated excess debt beta in } (r, m).$$

Then estimate:

$$fpr_{r,m} = \eta_0 + \eta_1 \beta_{r,m} + \varepsilon_{r,m}. \quad (11)$$

The paper reports a strong positive η_1 and a large R^2 (equation (12) in the paper).

(iv) Industry premiums outside finance. For non-financial industries i , compute an industry premium ip_i and an excess debt beta β_i and regress:

$$ip_i = \eta_0 + \eta_1 \beta_i + \varepsilon_i, \quad (13)$$

which is again strongly positive in the data.

6 Interpretation of all Figures (1–6, B1)

Figure 1: Time series of the financial premium

What it is: Monthly estimates of ϕ from a within-month version of (1/9).

What I learn:

- The premium is typically positive and spikes sharply during major stress periods (especially 2008–2009).
- The premium is countercyclical (higher in recessions), consistent with a systematic-risk interpretation.

Figure 2: Distribution of bond maturity and rating (financial vs industrial)

What it is: Histograms comparing the rating distribution and maturity distribution across issuer types.

What I learn:

- Financial issuers are, on average, higher rated and shorter maturity.
- This directly motivates the tight matching (month $\times$ rating $\times$ maturity FE) used to identify ϕ .

Figure 3: Financial premium vs excess debt beta

What it is: A scatter across rating–maturity subsamples of (excess beta, premium).

What I learn:

- There is a strong positive relationship: subsamples where financial bonds have higher beta differences also display higher premiums.
- This is precisely the model mechanism: the premium is compensation for systematic exposure.

Figure 4: Premium with transaction-based liquidity controls

What it is: Monthly premium series estimated with (and without) transaction-based liquidity measures (trades, large trades, volume, roundtrip costs), in the TRACE era.

What I learn:

- The two series nearly coincide.
- Liquidity differences are therefore unlikely to be the main driver of the financial premium.

Figure 5: Premium with covenant controls

What it is: Monthly premium series including indicators for a large set of covenants as controls.

What I learn:

- The premium remains similar with covenant controls.

- Therefore, contract-feature differences (as proxied by recorded covenants) do not explain the premium.

Figure 6: Premium adjusted for potentially slow rating updates

What it is: The premium series using only observations where the rating is unchanged six months ahead.

What I learn:

- The premium is still present under this restriction.
- Hence the result is not an artifact of financial-firm ratings being systematically “stale” relative to industrial firms.

Figure B1: Random recovery robustness

What it is: The model-implied $f(p, R, T)$ as a function of default probability p under fixed recovery versus random recovery (matched to the same expected recovery).

What I learn:

- The financial premium is robust to allowing recovery to be stochastic (quantitatively similar curves).

7 Interpretation of all Tables (1–10)

Table 1: Bond sample statistics

What it reports: Summary statistics for all bonds and separately for financial and industrial bonds, split by rating classes.

Main interpretation:

- The sample contains thousands of issuers and hundreds of thousands of bond-month observations.
- Within each rating class, financial spreads are higher than industrial spreads (the core fact).
- Financial firms tend to have higher ratings and shorter maturities, motivating careful matching.

Table 2: Most common issuers

What it reports: The issuers with the largest number of bond-month observations.

Main interpretation:

- The financial side is dominated by large banks/investment banks and finance subsidiaries of industrial conglomerates (e.g., captive finance).
- The industrial side includes large household-name corporates.
- This helps interpret the sample composition and why controlling for issuer-type characteristics matters.

Table 3: The financial premium across periods, ratings, and maturities

What it reports: Estimates of ϕ from (1/9) across subsamples, plus the premium as a fraction of average spreads.

Main interpretation:

- The premium is economically large and statistically significant in the full sample.
- It is larger in the financial crisis period, and larger for riskier ratings.
- It remains positive outside the crisis, so it is not a pure “crisis-only” phenomenon.

Table 4: Industry premiums (broad industries)

What it reports: Pairwise industry premiums estimated within rating–maturity–month cells.

Main interpretation:

- Finance stands out: the typical finance-versus-other premium is much larger than typical differences across non-financial industries.
- This supports the claim that the premium is a distinctive feature of financial debt, not generic industry segmentation.

Table 5: Financial sub-industries

What it reports: Premiums across sub-industries within finance (banking, credit/financing, financial services, insurance).

Main interpretation:

- Differences within finance are comparatively small (much smaller than finance vs non-finance).
- This fits the model logic: as long as the financial firm holds a large diversified debt portfolio, the exact asset composition is second-order for the premium.

Table 6: Financial excess debt betas

What it reports: Estimates of η from equation (10): financial-minus-industrial bond beta differences within rating-maturity-month cells.

Main interpretation:

- Financial bonds exhibit systematically higher betas than comparable industrial bonds.
- The excess beta is typically larger for riskier ratings and longer maturities, consistent with the model’s comparative statics.

Table 7: Default and recovery adjustments

What it reports: Premium estimates using unadjusted, liquidity-adjusted, and *loss-adjusted* spreads (with alternative Sharpe-ratio assumptions).

Loss-adjusted spread model:

$$\tilde{s}_{itj} = s_{itj} - \left(-\frac{1}{T_{it}} \right) \log \left(1 - (1 - \delta_{jt}) \Phi \left[\Phi^{-1}(\pi_{j,tT}^{\mathbb{P}}) + s^i \sqrt{T_{it}} \right] \right), \quad (14)$$

$$\tilde{s}_{itj} = s_{itj} - \left(-\frac{1}{T_{it}} \right) \log \left(1 - (1 - \delta_{jt}) \pi_{j,tT}^{\mathbb{P}} \right) \quad \text{if } s^i = 0. \quad (15)$$

Main interpretation:

- The premium remains large after loss adjustments, so it is not explained by financial bonds having higher expected losses conditional on rating.
- Hence, the evidence points toward a *risk-premium* explanation rather than a *default-probability mismeasurement* explanation.

Table 8: Transaction-based liquidity statistics

What it reports: Trading frequency, volume, and roundtrip costs in TRACE-era data.

Main interpretation:

- The “typical” (median) financial bond is not necessarily more liquid; transaction costs can be higher.
- Together with Figure 4, this supports the conclusion that liquidity is not the main driver of the premium.

Table 9: Systemic risk measures and beta

What it reports:

- Panel A: correlations between the financial premium time series and systemic risk proxies (SRISK, VIX, etc.).
- Panel B: regressions of portfolio-level premiums on systemic-importance indicators and excess beta.

Main interpretation:

- The premium comoves with systemic risk measures, consistent with the idea that it is compensation for systematic exposure.
- Systemic-importance indicators partly predict a larger premium, but their effect weakens once beta is controlled for, suggesting beta is a key sufficient statistic for the priced component.

Table 10: Large issuers and demand-curve effects

What it reports: The premium ϕ_1 estimated jointly with controls for issuer size (issuance and amount outstanding):

$$s_{itj} = \phi_1 \mathbf{1}\{j \in \text{Financial}\} + \phi_2 Y_{it} + \gamma' X_{it} + \mu_{mrt} + \varepsilon_{itj}.$$

Main interpretation:

- Large issuers can face yield effects (price pressure / demand curve).
- Yet the estimated financial premium remains remarkably stable across these specifications.
- Therefore, the premium is not an artifact of financial firms being systematically “large issuers.”

8 Conclusion

The paper establishes a robust and economically meaningful *financial premium* in corporate bond markets: even after controlling carefully for rating and maturity, financial issuers pay higher spreads than industrial issuers. A parsimonious model explains this fact by emphasizing that financial firms hold large diversified portfolios of non-financial debt, which concentrates their asset-side risk exposure on systematic factors; consequently, their liabilities are priced with higher systematic-risk compensation than standalone non-financial debt at the same default probability.

The empirical results line up tightly with the mechanism: financial bonds have higher betas, and the premium is increasing in the beta gap. Extensive robustness checks reject alternative explanations

based on mismeasured default losses, liquidity, covenants, large-issuer effects, or rating sluggishness. Conceptually, the financial premium is best interpreted as a *systematic-risk premium embedded in financial-sector credit*, not a statistical artifact of bond characteristics or rating practices.

A Appendix A: Derivation and proof skeletons (core equations)

This appendix rewrites the main derivations in a self-contained way, following the paper's Appendix A.

A.1 A.1 Pricing firm debt (Merton with exogenous recovery)

Under the risk-neutral measure $\mathbb{Q}$, the time-0 price per unit face value of firm debt is

$$B_i(0) = \mathbb{E}^{\mathbb{Q}} \left[e^{-rT} \left(\mathbf{1}\{V_i(T) > D_f\} + R_f \mathbf{1}\{V_i(T) < D_f\} \right) \right]. \quad (16)$$

Using lognormality, this becomes

$$B_i(0) = e^{-rT} \left(1 - (1 - R_f) \Phi(-d_2(r, \sigma, T, D_f, V_i(0))) \right), \quad (17)$$

where

$$d_2(r, \sigma, T, D_f, V_i(0)) = \frac{\log(V_i(0)/D_f) + (r - \sigma^2/2)T}{\sigma\sqrt{T}}. \quad (18)$$

Thus the risk-neutral default probability is

$$\mathbb{Q}(V_i(T) < D_f) = \Phi(-d_2(r, \sigma, T, D_f, V_i(0))). \quad (19)$$

The physical default probability is

$$\mathbb{P}(V_i(T) < D_f) = \Phi(-d_2(\mu, \sigma, T, D_f, V_i(0))). \quad (20)$$

Solving $\mathbb{P}(V_i(T) < D_f) = p_f$ pins down the face value:

$$D_f(p_f) = V_0 \exp \left(-\sigma\sqrt{T}\Phi^{-1}(p_f) + (\mu - \sigma^2/2)T \right). \quad (21)$$

A.2 A.2 Mapping p_f to the risk-neutral default probability $q_f(p_f)$

Define the firm asset Sharpe ratio $s_f = (\mu - r)/\sigma$. Then

$$q_f(p_f) = \Phi \left(\Phi^{-1}(p_f) + s_f\sqrt{T} \right). \quad (22)$$

In the CAPM environment of the paper, $s_f = \sqrt{\rho} s_m$, which yields the argument in Proposition 1.

A.3 A.3 Diversified loan portfolio and bank default probability

Let F denote the (large- N) fraction of firm loans that default. Under the Vasicek approximation:

$$\mathbb{P}(F \leq x) = \Phi\left(\frac{1}{\sqrt{\rho}}\left(\sqrt{1-\rho}\Phi^{-1}(x) - \Phi^{-1}(p_f)\right)\right), \quad (23)$$

$$\mathbb{Q}(F \leq x) = \Phi\left(\frac{1}{\sqrt{\rho}}\left(\sqrt{1-\rho}\Phi^{-1}(x) - \Phi^{-1}(q_f(p_f))\right)\right). \quad (24)$$

If the bank chooses leverage (face value) D_b to target default probability p_b , then D_b solves

$$p_b = \mathbb{P}\left(F > 1 - \frac{D_b}{1 - R_f}\right), \quad (25)$$

which yields

$$D_b(p_b, p_f, R_f) = 1 - (1 - R_f)\Phi\left(\frac{\Phi^{-1}(p_f) - \sqrt{\rho}\Phi^{-1}(p_b)}{\sqrt{1-\rho}}\right). \quad (26)$$

The corresponding risk-neutral default probability of the bank is

$$q_b(p_b, p_f, R_f) = \Phi\left(\Phi^{-1}(p_b) + s_m\sqrt{T}\right). \quad (27)$$

This shows why the bank spread depends on p_b, R_b, T and s_m , but not on the idiosyncratic-risk components of individual loans once the portfolio is well diversified.

A.4 A.4 Debt betas (sketch)

Debt beta can be written as the ratio of market-exposure volatility to market volatility:

$$\gamma_d(t) = \frac{\sigma_{dm}(t)}{\sigma_m}, \quad (29)$$

and applying Itô to the debt price yields for firm debt:

$$\gamma_f(t) = \frac{B_V^f(t)V_i(t)}{B^f(t)}\beta, \quad (30)$$

which, after differentiating the Merton price and substituting in the default-probability mapping, gives equation (6). A parallel argument for bank debt (where only the market factor remains after diversification) gives equation (7).

B Appendix B: All original tables and figures from the paper

To satisfy “include all tables”, I embed the paper’s pages that contain Figures and Tables. (These pages are included verbatim from the PDF to preserve exact formatting.)